Electric car: launch the design

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Plan a safe team workflow and identify the car's energy pathway.

SUCCESS CRITERIA - I can...

1. Label the battery, switch, motor and connections.
2. Identify safety checkpoints before tools or power.
3. Connect a design choice with a measurable outcome.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use "prototype" and "constraint" to explain your model or result. Sentence starters are optional.

prototype: a version built to test a design

constraint: a requirement or limit that the design must meet

evidence: observations or measurements that support an idea

01 Warm-up: bring back an earlier idea

A single-loop toy calculation gives 3 V induced emf. **Explain** why this alone cannot predict the effects on a whole power grid; name two missing features.

Sentence starter: I predict … . My model supports this because … .

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: average speed = distance/elapsed time. Use your own design and measurements. Battery → motor → drivetrain. Follow teacher training for tools and hot materials.

Worked reasoning: A design claim becomes testable when it names a measurement. "The car is fast" can become "The car travels 3 m in under 2 s on this level track." That condition corresponds to average speed above 1.5 m/s.

Your turn: Draw a test track with start, finish, distance and timing method labeled.

Sentence starter: The … in my model represents … . Its label shows … .

Start Finish



Distance: ____ Timing method: _____

Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

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Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

Sketch your proposed car.
Label the battery, switch, motor, driven axle and wheels.
Add a theme that will not obstruct moving parts.

Sentence starter: The ... in my model represents ... . Its label shows ... .



Top view: the large rectangle is the chassis; small rectangles are wheels; the two crossbars are axles.

Marzano check - circle: 1 2 3

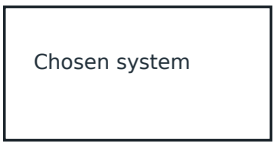
04 Apply the physics

Your first design uses a 3 V supply and an expected motor current of 0.40 A.
Estimate electrical input power.
Label at least three destinations that prevent all input power from becoming useful forward-motion power.

Word help: estimate = find an approximate value

Sentence starter: I predict ... . My model supports this because ... .

Draw and label all energy transfers across the boundary.



Stored energy change: _____ J

Marzano check - circle: 1 2 3

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

Write one measurable design goal and one constraint. **Explain** how you will measure the goal instead of judging only by appearance.

Sentence starter: My design shows … . It meets the requirement by … .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A circuit has 4 identical series resistors. Add two more at unchanged ideal supply voltage.
Compare new current with the original.
Draw the stated circuit and label quantities relevant to the question.

Word help: current = rate of charge flow

Sentence starter: Both … . They differ in … because … .

Draw the complete circuit and label shared nodes.

Marzano check - circle: 1 2 3

07 Exit check

Create a fair first-test plan naming distance, timing method, number of trials and variables held fixed.

Sentence starter: My design shows … . It meets the requirement by … .

Check / criterion	Measurement	Decision rule
___	___	___
___	___	___

Marzano check - circle: 1 2 3

08 STRETCH: test the model

Choose a speed target and a reliability target for a 3 m course.
Design repeated trials and a rule for judging success without selecting only the fastest run.

Sentence starter: My design shows … . It meets the requirement by … .

Check / criterion	Measurement	Decision rule
___	___	___
___	___	___

Marzano check - circle: 1 2 3